# Optimal Packing of Fifteen Equal Disks in a Circle: An Exact Computer-Assisted and Lean-Verified Proof

October 12, 2026

**Abstract**

We prove that the minimum radius of a circle containing fifteen pairwise interior-disjoint unit disks is

$$R_0 = 1 + \sqrt{1 + \big(2 + \cot(\pi/5)\big)^2} = 4.521356964706164\ldots$$

An exact fivefold-symmetric construction gives the upper bound. For the lower bound, a radial threshold reduces every putative packing with radius at most $R_0$ to four possible inner/outer counts. For each of the 760 dihedral classes of their cyclic binary words, an integer difference-constraint test either excludes the class or passes it to exact rational interval refinement. Refinement leaves one cyclic type, $(I, O, O)^5$, and confines its five inner-center radii to $[1.68, 1.72]$. An explicit local angular inequality then forces the candidate radius vector and excludes every smaller container. The certificate uses integer and rational arithmetic, including an explicit Taylor remainder bound for the trigonometric table. A Lean theorem now characterizes feasibility exactly by $R \geq R_0$; the finite replay and its trust dependencies are described below.

**Verification status.** The repository includes a serialized subdivision certificate, a separate rational checker, and a one-command replay. The Lean project also constructs the feasible candidate packing and proves the matching lower bound for arbitrary unit-disk packings, yielding an exact feasibility characterization. The finite replays use `native_decide`; thus the theorem is formalized in Lean with explicit native-computation trust dependencies, rather than being kernel-only. The scope and trust boundary are recorded in section B.

## 1 Introduction

Packing congruent disks in a circle is equivalent to maximizing the minimum distance among a fixed number of points in a disk. Graham proved the small cases through seven points, and Pirl treated the circle-packing cases through ten. Later computational work by Graham, Lubachevsky, Nurmela, and Ostergård tabulated best-known configurations and described the configurations for 12 through 20 disks as conjecturally optimal; their 15-disk entry is the familiar five-inner, ten-outer arrangement [1, 2, 3]. Ekanayake and LaFountain subsequently proved the 14-disk case and reported that the proved cases then known were $N \leq 14$ and $N = 19$ [4]. The 15-disk case is therefore a natural next problem.

**Proof overview.** The main difficulty is to exclude nonsymmetric packings without assuming an optimizer's contact graph. We first replace each outer center by a wall center at the same polar angle, then convert pairwise separation into angular difference constraints. An exact finite certificate checks all 760 cyclic order classes and leaves one type, $(I, O, O)^5$, with five inner radii

in a small interval. In that interval, the direct inner-pair constraint and the route through the two intervening outer centers respond in opposite ways to radial perturbations. Their maximum creates a strict angular penalty away from the symmetric candidate. The finite stage rejects a radial box only when a necessary condition fails; a surviving box is not treated as evidence that a packing exists.

## 2 Problem and main result

Let $p_1, \dots, p_{15} \in \mathbb{R}^2$ be the centers of fifteen unit disks. A circle of radius $R$ centered at the origin contains the disks without interior overlap exactly when

$$\|p_i\| \le R - 1 \quad (1 \le i \le 15), \qquad \|p_i - p_j\| \ge 2 \quad (i \ne j). \tag{1}$$

Write $R_{15}^*$ for the infimum of feasible $R$.

**Theorem 2.1.** *The minimum enclosing radius for fifteen pairwise interior-disjoint unit disks is*

$$R_{15}^* = R_0 = 1 + \sqrt{1 + \big(2 + \cot(\pi/5)\big)^2}.$$

The construction in section 3 proves the upper bound, and the finite reduction and local barrier below prove the matching lower bound. The Lean development now states the same result as an exact characterization of all feasible container radii; its finite-computation trust dependencies are listed in section B.

The upper bound comes from the familiar $D_5$-symmetric arrangement, invariant under rotations by $2\pi/5$ and reflections, with five inner disks and ten outer disks; it is listed among the best-known configurations [3, 5]. The lower-bound strategy is independent of the tight-partition reduction used for fourteen disks [4].

## 3 An exact packing of radius $R_0$

Set

$$\varphi = \frac{\pi}{5}, \qquad a_0 = \csc\varphi, \qquad b_0 = \sqrt{1 + \big(2 + \cot\varphi\big)^2}, \qquad \alpha = \varphi - \arcsin(1/b_0), \qquad R_0 = 1 + b_0.$$

For $j = 0, \dots, 4$, take centers

$$I_j = (a_0, 2j\varphi), \qquad O_j^- = (b_0, 2j\varphi - \alpha), \qquad O_j^+ = (b_0, 2j\varphi + \alpha)$$

in polar coordinates. The five inner centers form a regular pentagon, with two outer centers between each pair of consecutive inner-center directions [5].

Adjacent inner centers have distance $2a_0 \sin\varphi = 2$. The smaller of the two successive angular gaps between outer centers is $2(\varphi - \alpha)$, and

$$2b_0 \sin(\varphi - \alpha) = 2.$$

The other outer gap is larger because $\alpha > \varphi/2$. For an inner center and either of its two nearest outer centers, the cosine rule gives

$$a_0^2 + b_0^2 - 2a_0 b_0 \cos\alpha = 4.$$

These are the closest pairs within each type; all other same-type or mixed pairs have a larger angular separation and hence a larger center distance. Every outer disk is tangent to the container boundary, and $a_0 < b_0$. Thus the arrangement is feasible in radius $R_0$, proving $R_{15}^* \le R_0$; figure 1 shows the resulting placement.

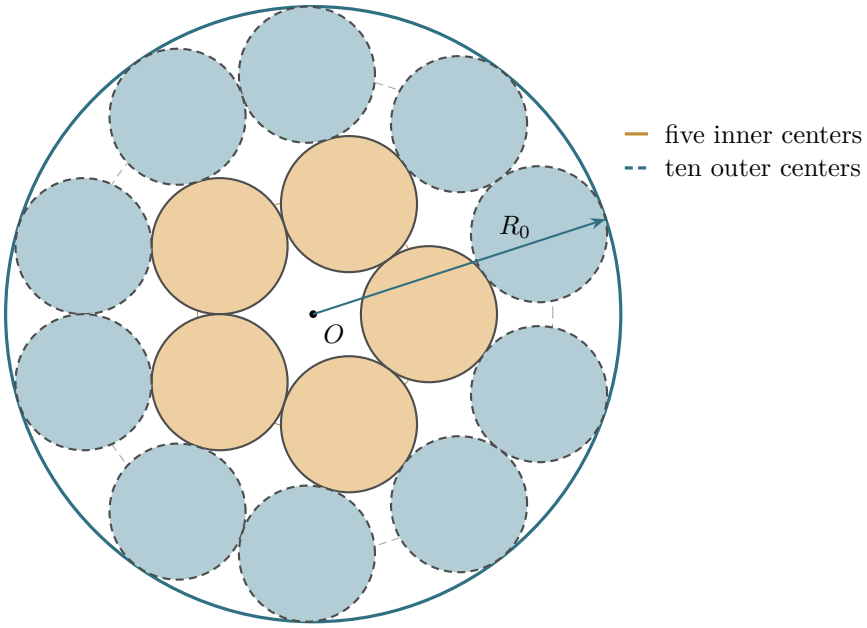



Figure 1: The exact fivefold-symmetric construction. Inner disks have solid outlines and outer disks dashed outlines, so the two types remain distinguishable in grayscale. The dashed concentric circles mark center radii. This verifies the upper bound only and makes no symmetry assumption about a minimizer.

## 4 Radial reduction

For a center at distance $s$ from the origin, call it inner if $s < L$ and outer if $s \geq L$, where

$$L = b_0 - \frac{4}{b_0}. \tag{2}$$

The decimal enclosures used by the verifier are rational outward bounds; in particular $2.385431 < L < 2.385432$ and $b_0 < 3.522$.

For $x, y > 0$ define

$$c(x,y) = \frac{x^2 + y^2 - 4}{2xy}, \qquad \Phi(x,y) = \arccos c(x,y)$$

whenever $|x - y| < 2 \leq x + y$. If the two center radii are $x, y$ and their smaller angular separation is $\Delta \in [0, \pi]$, non-overlap requires $\Delta \geq \Phi(x,y)$.

These are radial bookkeeping labels; they do not assume that an optimizer has five centers of one kind and ten of the other. The threshold is chosen so the mixed corner $(L, b_0)$ and the wall corner $(b_0, b_0)$ require the same angle:

$$c(L, b_0) = c(b_0, b_0).$$

This identity is the key to the outer–outer comparison in the wall-replacement lemma below.

**Lemma 4.1** (Four-corner maximum)**.** *On every positive rectangle $[u_1, u_2] \times [v_1, v_2]$, the maximum of $c(x,y)$ is attained at a corner.*

*Proof.* For fixed $y$,

$$\frac{\partial c}{\partial x} = \frac{x^2 - y^2 + 4}{2x^2 y}.$$

Its numerator is strictly increasing in $x$, so the derivative can change sign only from negative to positive. Thus an interior critical point in the $x$ direction is a minimum, and a maximum occurs at an endpoint. Apply the same argument in $y$. □

**Lemma 4.2** (Replacing outer centers by wall centers)**.** *Suppose a configuration satisfies (1) for some $R \le R_0$. Simultaneously replace every outer center of radius $r$ by a center of radius $b_0$ at the same polar angle. The resulting configuration satisfies all pairwise non-overlap constraints and is feasible in a container of radius $R_0$. This is a statement about the final configuration; it does not assert that a continuous motion between the two configurations stays feasible.*

*Proof.* For an inner–outer pair, let $x < L \le y \le b_0$ be the inner and outer radii before replacement. Its squared distance at fixed angular difference is increasing in the outer radius, since

$$\frac{d}{dy}\left(x^2 + y^2 - 2xy\cos\Delta\right) = 2(y - x\cos\Delta) > 0,$$

so the pair remains separated after the outer radius is replaced by $b_0$. For an outer–outer pair, lemma 4.1 and $L = b_0 - 4/b_0$ show that

$$\max_{L \le x, y \le b_0} c(x, y) = c(b_0, b_0),$$

with equality also at $(L, b_0)$ and $(b_0, L)$. Feasibility before replacement gives $\cos\Delta \le c(x, y) \le c(b_0, b_0)$, so this pair is also separated in the resulting configuration. □

Let $\theta_0 = 2\arcsin(1/b_0)$. Any two outer centers, after the preceding move, require angular separation at least $\theta_0$. Since $11\theta_0 > 2\pi$, there are at most ten outer centers. Nine inner centers would be nine unit disks inside radius $L + 1 < 3.4$, whereas the exact nine-disk minimum radius is $1 + \sqrt{2(2 + \sqrt{2})} > 3.6$ [2, 5]; hence there are at most eight inner centers. Since there are fifteen disks in all,

$$(|\mathcal{I}|, |\mathcal{O}|) \in \{(5, 10), (6, 9), (7, 8), (8, 7)\}. \tag{3}$$

The threshold and the surviving count ranges are sketched in figure 2.

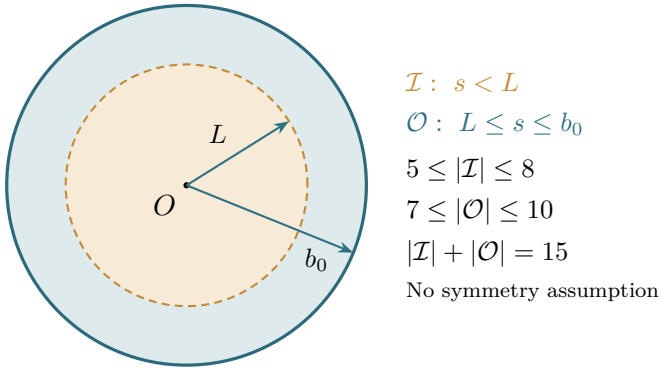



Figure 2: The radial threshold separates inner centers from the outer annulus. The count bounds leave only the four populations in equation (3); this is not a packing diagram.

Write $s_{(1)} \le \cdots \le s_{(15)}$ for the fifteen center radii in nondecreasing order. The standard small-packing bounds $s_{(5)} \ge a_0 > 5/3$ and $s_{(6)} \ge 2$ [1] are also used in the radial interval enumeration. They follow by applying the known exact five- and six-disk packing radii to the five and six innermost centers. Consequently, among $k$ inner centers at least $k-4$ have radius at least $5/3$, and at least $k-5$ have radius at least 2.

## 5 Angular difference constraints

Order the centers by nondecreasing polar angle and lift their angles to $\theta_1 \le \cdots \le \theta_{15} < \theta_1 + 2\pi$. For a pair whose radii lie in positive intervals $X_i, X_j$, let

$$C_{ij} = \max\{c(x,y) : x \in X_i,\ y \in X_j\}.$$

The maximum is obtained at the four interval corners by lemma 4.1. The exact zero-radius case and positive intervals whose closure touches zero are handled separately; the endpoint rules and their bin ownership are proved in section 6.1.

If the interval box gives $x+y < 2$ for every choice of radii in a pair, that pair cannot be separated, so the box is discarded. For every pair not discarded by this test, $C_{ij} \ge -1$. If $-1 \le C_{ij} < 1$, put $\gamma_{ij} = \arccos C_{ij}$. Every feasible angle assignment must satisfy

$$\gamma_{ij} \le \theta_j - \theta_i \le 2\pi - \gamma_{ij} \qquad (i < j). \tag{4}$$

A box for which $|x-y| \ge 2$ throughout imposes no angular restriction for that pair. These rules are relaxed outwards, so every actual packing in a radius box satisfies the resulting difference constraints. The geometric conversion is illustrated in figure 3.

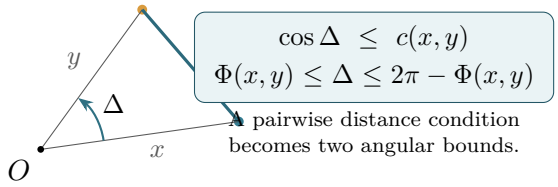



Figure 3: The law of cosines converts center separation into a lower bound on the smaller angular gap. The complementary upper bound in equation (4) is also needed for the difference-constraint graph.

Each inequality $\theta_v - \theta_u \le w$ is a directed edge $u \to v$ with weight $w$. Such a system is feasible only if the directed graph has no negative cycle. The verifier replaces each required angle by a certified rational lower bound $q/2800$ and uses $44/7$ as an upper bound for $2\pi$. Thus a lower-bound edge has weight $-q$, while its complementary upper-bound edge has weight $17600 - q$. A negative cycle in these integer weights excludes the entire radial box.

For example, suppose three angle constraints require

$$\theta_2 - \theta_1 \ge \lambda, \qquad \theta_3 - \theta_2 \ge \mu, \qquad \theta_3 - \theta_1 \le \nu.$$

The first two imply $\theta_3 - \theta_1 \ge \lambda + \mu$, so no angles can satisfy all three if $\lambda + \mu > \nu$. In the difference graph, these inequalities give a directed cycle of total weight $\nu - \mu - \lambda < 0$.

**Certified trigonometric bounds.** For $x \geq 0$, let $P_{18}(x) = \sum_{j=0}^{9} (-1)^j x^{2j}/(2j)!$. Taylor's theorem and $|\cos^{(19)}(t)| \leq 1$ give the rational lower bound

$$\cos x \geq P_{18}(x) - \frac{x^{19}}{19!}. \tag{5}$$

The explicit remainder subtraction is essential: the even Taylor polynomial alone is not used as a lower bound. For an interval pair with corner maximum $C_{ij}$, the integer $q$ is chosen so that the right side of (5) at $x = q/2800$ is at least $C_{ij}$. Since cosine decreases on $[0, \pi]$, this certifies $q/2800 \leq \gamma_{ij}$. The integer range is certified by exact Machin-series bounds: $8790/2800 < \pi < 22/7$, so every tested angle is below $\pi$ and $2\pi < 44/7$. All comparisons are performed with exact fractions.

## 6 Exhaustive finite certificate

The cyclic inner/outer word is defined up to rotation and reflection, because the disks are congruent and the container is rotationally and reflectively symmetric. Burnside's lemma gives 111, 185, 232, and 232 classes for the four inner counts in equation (3), for a total of 760. The word records only the order of the two radial labels around the circle; it does not impose geometric symmetry, as illustrated in figure 4.

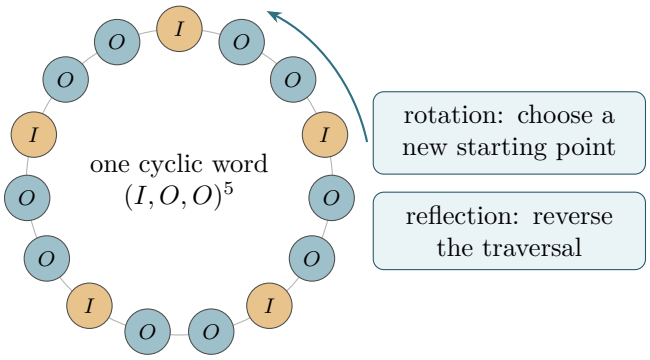



Figure 4: A binary bracelet records only the cyclic inner/outer order. Rotating or reversing the word changes its representation, not the underlying case; it does not make a geometric packing symmetric.

**Proposition 6.1** (Finite reduction)**.** *After replacing outer centers by wall centers, every feasible packing of radius at most $R_0$ has cyclic type $(I, O, O)^5$, and its five inner-center radii lie in*

$$[42/25, 43/25]^5 = [1.68, 1.72]^5.$$

*This is the conclusion of the finite certificate. A terminal box contained in this cube is passed to the analytic argument; it is not asserted to contain a feasible packing.*

The finite certificate for proposition 6.1 has three stages. First, exact rational arithmetic checks a fixed four-type angular table, and an integer negative-cycle test examines all 760 words. Second, the surviving words are refined over eight radial bins and then twelve radial bins, using the rank bounds in section 4. Third, exact rational subdivision excludes each remaining radial box by a necessary condition, except that the canonical type is retained when a box lies inside the local cube. The finite computation ends there; the local analytic lemma in the next section rules out every point of that cube except the candidate radius vector.

Here a box is called closed only when the entire represented radial region is excluded by a stated necessary condition. A local leaf records a box passed to the analytic proof and makes no existence claim. The flow in figure 5 summarizes the three finite stages and the subsequent local argument.

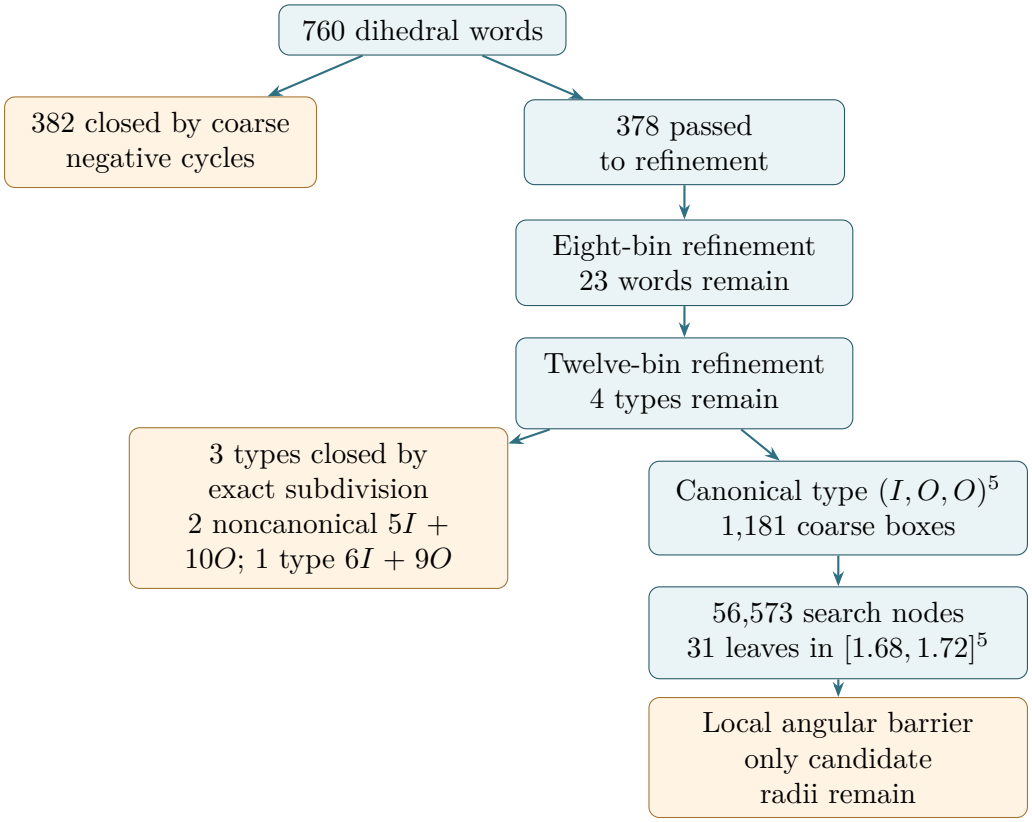



Figure 5: The first three stages form the finite reduction and test all 760 dihedral classes. The final analytic step handles the boxes passed to local rigidity.

The exact rational subdivision closes the two noncanonical $5I+10O$ words from 47 and 38 coarse radial boxes, respectively. Their branch-and-bound trees contain 1,641 and 646 nodes. The $6I+9O$ word has three coarse boxes. Splitting its unique first-bin radius into $0 < r < 1/4$, $1/4 \le r < 1/2$, and the exact origin gives nine leaves: six pair-sum contradictions and three negative cycles.

For the canonical word $(I, O, O)^5$, 1,181 coarse radial boxes generate 56,573 search nodes: 28,846 negative-cycle leaves, no pair-sum leaves, and 31 leaves passed to the local argument. The local proof below excludes every point in those boxes except the candidate radius vector, without assuming that the five radii or the packing are symmetric.

The endpoint ownership convention, origin case, and closed-hull bounds are recorded in section 6.1. The serialized data and independent certificate checks are described in the reproducibility appendix.

### 6.1 Endpoint ownership and the origin

The finite enumerators assign each radius to a half-open owner bin: every rational cutpoint belongs to the bin on its right. The first owner bin is split into the singleton $\{0\}$ and the positive interval $0 < r < 1/2$. The remaining owner bins begin at $1/2$, 1, $3/2$, $8/5$, $5/3$, and 2 (with the stated further subdivisions in the twelve-bin stage); the last inner owner is truncated at $L$. The arrays

used for interval bounds are closed rational hulls of these owner bins. Consequently, the outward bounds may include shared endpoints, but the discrete enumeration neither loses an endpoint nor relies on assigning it to two different cases. At an exact bisection, the two closed child hulls cover the parent, including their common midpoint.

This convention justifies the early count tests. The labels counted by 'small' have actual radii strictly below 1, so 'small>1' gives two centers with radial sum strictly less than 2. A first-bin center and a center in $[1, 3/2)$ also have sum strictly less than 2. The labels counted by 'high' and 'high2' have lower endpoints $5/3$ and 2, respectively, so the rank bounds in the radial reduction are preserved at equality.

An exact-origin center has no polar angle. The graph therefore places no angular edges between it and other centers. Instead, since its distance to a center of radius $s$ is exactly $s$, every other center must have radius at least 2. The refined enumerators enforce this directly: all other inner labels must be in bins with lower endpoint at least 2 (label 6 in the eight-bin grid and label 8 in the twelve-bin grid); the wall-pushed outer centers already have radius $b_0 > 2$.

For the positive first-bin type, $0 < x < 1/2$, the closed hull touches zero but does not represent the origin. When the other interval has upper endpoint at most 2, the cosine expression $c(x, y)$ is increasing in each variable on the relevant positive rectangle, since

$$\partial_x c = \frac{x^2 - y^2 + 4}{2x^2 y} \geq 0, \qquad \partial_y c = \frac{y^2 - x^2 + 4}{2xy^2} \geq 0.$$

Thus the value at the upper endpoints is a valid upper bound even as $x \downarrow 0$. If the other interval extends above 2, the verifier disables that angular edge. The exact-origin singleton is never subjected to this limiting-angle rule.

## 7 The local angular barrier

Assume the cyclic word is $(I, O, O)^5$. Let $a_i$ be the five inner radii in cyclic order, with indices modulo five. Every outer radius is at most $b_0$.

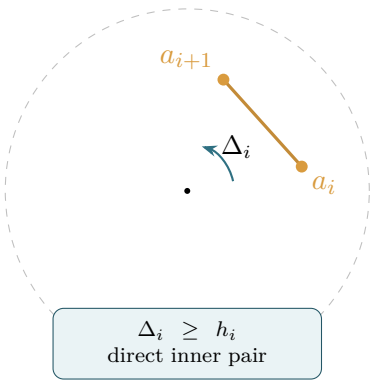
For each sector between consecutive inner-center directions, there are two independent routes that force it to use angular space. The direct chord between the inner centers gives one lower bound. The path through the two intervening outer centers gives another. Increasing the inner radii relaxes the direct inner-pair condition, but tightens the route through the outer centers; decreasing them reverses the roles. We take the larger bound so that either direction of radial change incurs an angular cost. The quantities below encode these two routes:

$$h_i = \Phi(a_i, a_{i+1}), \qquad k_i = \Phi(a_i, b_0) + \Phi(b_0, b_0) + \Phi(a_{i+1}, b_0).$$

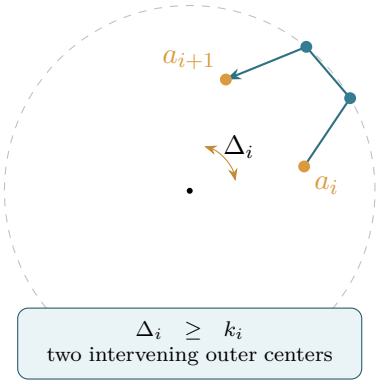
For $a_i \in [1.68, 1.72]$, the exact interval checks give $k_i > 1.175$. The four complementary inner-to-inner gaps therefore sum to more than $4.7 > \pi$, so the remaining gap is less than $\pi$. The direct inner pair and the path through the two intervening outer centers then give a lower bound $\max(h_i, k_i)$ on that gap. Summing the five gaps yields the necessary condition

$$\sum_{i=1}^{5} \max(h_i, k_i) \leq 2\pi. \tag{6}$$

When an outer center has radius below $b_0$, the inner–outer angle requirement is at least the value used in $k_i$, because $\Phi(a, s)$ decreases with $s$ on the relevant interval. The outer–outer term obeys the analogous bound.




(a) The inner chord requires at least $h_i$.




(b) The three consecutive gaps sum to at least $k_i$.

Figure 6: Each inner-to-inner sector has two independent lower bounds. Combining them gives $\Delta_i \geq \max\{h_i, k_i\}$, which leads to the total angular budget in equation (6). The radii and angles are schematic.

**Lemma 7.1** (Local angular rigidity)**.** *If $a_i \in [1.68, 1.72]$ for every $i$ and the angular budget (6) holds, then $a_i = a_0$ for all $i$.*

*Proof.* At $a_i = a_0$, one has $h_i = k_i = 2\pi/5$. Write $\delta_i = a_i - a_0$ and $\sigma_i = \delta_i + \delta_{i+1}$. The direct route has first-order change $-c_0\sigma_i$, while the route through the two outer centers has first-order change $d_0\sigma_i$. Differentiation at the candidate gives

$$\partial_x \Phi(x,y)|_{(a_0,a_0)} = -c_0, \qquad \frac{d}{dx}\Phi(x,b_0)\Big|_{x=a_0} = d_0,$$

where

$$c_0 = \frac{\sin^2(\pi/5)}{\cos(\pi/5)} > \frac{21}{50}, \qquad d_0 = \cos(\pi/5) > \frac{1}{2} > c_0.$$

The second derivatives on $[1.68, 1.72]^2$ and on $[1.68, 1.72] \times \{b_0\}$ are bounded, with exact rational checks, by

$$|\Phi_{xx}(x,y)| < 0.3, \qquad |\Phi_{xy}(x,y)| < 0.8, \qquad |\Phi_{xx}(x,b_0)| < 5.$$

Taylor's theorem therefore gives

$$h_i \geq \frac{2\pi}{5} - c_0\sigma_i - \frac{5}{2}(\delta_i^2 + \delta_{i+1}^2),$$
$$k_i \geq \frac{2\pi}{5} + d_0\sigma_i - \frac{5}{2}(\delta_i^2 + \delta_{i+1}^2).$$

Since $d_0 > c_0 > 0$, $\max\{-c_0\sigma_i, d_0\sigma_i\} \geq c_0|\sigma_i|$: when one route becomes less restrictive, the other becomes more restrictive. The odd cycle length is essential. The linear map from the radial deviations $\delta_i$ to their consecutive sums $\sigma_i$ is invertible; an even cycle would admit a nonzero alternating mode in its kernel. Indeed, the following identity holds:

$$2\delta_i = \sigma_i - \sigma_{i+1} + \sigma_{i+2} - \sigma_{i+3} + \sigma_{i+4}.$$

Taking absolute values and summing over all five indices gives

$$\sum_i |\sigma_i| \geq \frac{2}{5}\sum_i |\delta_i|.$$

Moreover $|\delta_i| \le 11/500$ throughout the local cube. If at least one $\delta_i$ is nonzero, the final inequality below is strict because $\frac{2}{5}c_0 - 5 \cdot \frac{11}{500} > \frac{29}{500}$:

$$\begin{aligned}\sum_i \max(h_i, k_i) &\ge 2\pi + c_0 \sum_i |\sigma_i| - 5 \sum_i \delta_i^2 \\ &\ge 2\pi + \left(\frac{2}{5}c_0 - 5 \cdot \frac{11}{500}\right) \sum_i |\delta_i| \\ &> 2\pi + \frac{29}{500} \sum_i |\delta_i|.\end{aligned}$$

This contradicts (6) unless every $\delta_i = 0$. □




(a) The five radii vary independently.

(b) Taking the larger linear term yields a positive penalty.

Figure 7: The local barrier has no hidden symmetry constraint: radial deviations are independent, while the maximum of the oppositely sloped first-order terms grows away from $\sigma_i = 0$. Quadratic errors are controlled by the exact interval bounds.

If every $a_i = a_0$ but $R < R_0$, each original outer radius is strictly below $b_0$. Its required inner–outer angle is then strictly larger than the candidate value, so the five-gap sum is strictly larger than $2\pi$, again a contradiction. Thus no packing with $R < R_0$ exists in the final surviving class. Combining the explicit construction, the finite reduction, and the local angular rigidity lemma completes the proof of theorem 2.1.

## 8 Conclusion and reproducibility

The explicit construction proves $R_{15}^* \le R_0$, while the radial classification, exhaustive difference-constraint certificate, exact rational subdivision, and local angular barrier prove that every feasible packing has $R \ge R_0$. The Lean theorem `fifteen_unit_packing_radius_iff_ge_closed_form` packages both directions as an equivalence: a packing of fifteen unit disks exists in radius $R$ if and only if $R_0 \le R$.

Replay files are in the repository's candidate bundle pinned to commit `02b6176`. Run it with Python 3.10 or newer, GNU C++17, and a POSIX shell:

```
cd problems/packing-15-equal-disks-in-a-circle/candidate_bundle
bash run_all.sh
```

All exclusion decisions use integers or rational fractions. The code's decimal output is diagnostic only. The replay is self-contained and records the branch counts and terminal reasons.

The formal statement does not assume a contact graph or symmetry. Its finite certificate checks are replayed in Lean and by the separate rational checker; the trust boundary of the Lean replay is stated explicitly in the following section.

# A Certificate data and exact subdivision

The repository's candidate bundle pinned to commit `02b6176` contains the tables, branch rules, reports, and replay scripts. Stage zero writes `stage0_certificate.json.gz`, covering every excluded binary dihedral class and each radial type assignment not ruled out by rank constraints.

The independent `check_stage0_certificate.py` regenerates all 760 classes and checks assignment coverage and integer cycles. The replay also emits a 3.8 MB subdivision certificate with 1,266 roots. The independent `check_exact_certificate.py` verifies every bisection cover, pair-sum leaf, negative-cycle edge and weight, and local-region leaf using rational arithmetic; it checks the three $6I+9O$ assignments against `full_residuals.txt` and verifies all nine endpoint leaves. Stage counts and survivor identities are asserted during replay.

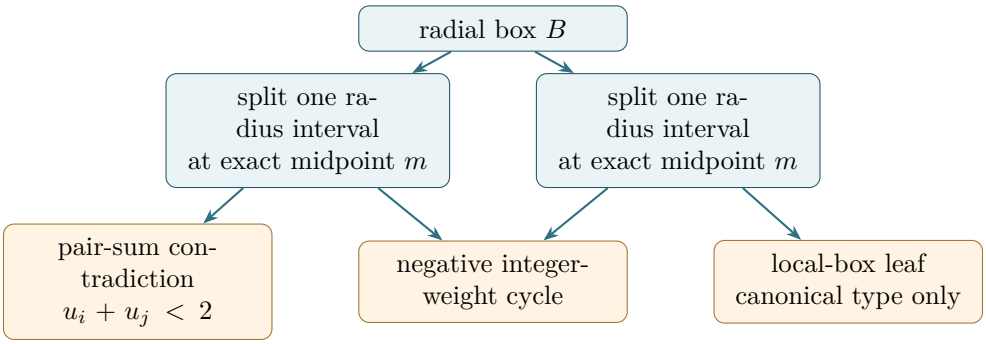



Figure 8: Each exact bisection covers its parent by two closed rational children, including the midpoint. A branch ends only at an exact contradiction or, for the canonical type, at a box passed to the separate local proof.

# B Lean formalization and trust boundary

The Lean development is available in the repository's Lean project pinned to commit `5a64563`. Although hosted under the unequal-disk problem directory, its `Fifteen` modules formalize the equal-disk problem in this manuscript. They use the same candidate radii $a_0, b_0$ and local cube $[42/25, 43/25]^5$.

| Manuscript component | Lean status at the pinned snapshot |
|---|---|
| Candidate construction (section 3) | `fifteenCandidatePacking` constructs a `Packing 15` at radius $R_0$; exact containment and all pairwise separation conditions are proved. |
| Radial reduction (section 4) | The arbitrary-packing reduction, wall replacement, radius classification, cyclic normalization, rank bounds, and interval ownership are connected to the finite Stage 0–4 checks used by the lower-bound theorem. |
| Finite reduction (proposition 6.1) | Certificate checks and their geometric soundness are applied to arbitrary packings; surviving canonical boxes are passed to the local argument. Surviving boxes are not treated as evidence of existence. |
| Local angular barrier (section 7) | Taylor error estimates, the odd five-cycle identity, and local rigidity are used in the strict smaller-container exclusion, including the wall-radius comparison for the original packing. |
| Optimality (theorem 2.1) | `fifteen_unit_packing_radius_iff_ge_closed_form` proves that a radius $R$ admits fifteen unit disks exactly when $1 + \sqrt{1 + (2 + \cot(\pi/5))^2} \le R$. |

**The final theorem.** The upper-bound witness is `fifteenCandidatePacking` in `CirclePacking/FifteenCandidatePacking.lean`. Its cyclic list uses five inner centers and ten outer centers; exact contact-angle identities and the cosine rule establish all 105 unordered pair constraints, and a radial inequality establishes containment. The lower-bound theorem `fifteen_unit_packing_stage4_strict_wall_impossible`, in `CirclePacking/FifteenStage4StrictCycle.lean`, starts from an arbitrary `Packing 15 R` with unit radii and proves impossibility when $R < R_0$. The theorem `fifteen_unit_packing_radius_iff_ge_closed_form`, in `CirclePacking/FifteenOptimality.lean`, combines these directions and states the exact value in the same closed form as theorem 2.1. The default `CirclePacking` target imports this theorem.

**Alternative intermediate proofs.** The Lean development sometimes proves the same required bound by a different route. For the eight-inner-center upper bound, it uses a contact-cosine bound $c(x, y) \le 2/3$ on $[1, 12/5]^2$ and $\arccos(2/3) > \pi/4$, together with the fact that at most one center has radius below 1. It therefore does not need the nine-disk optimum invoked in section 4. The 760 class counts are checked by exhaustive canonical-mask enumeration rather than by formalizing Burnside's formula. For local Taylor errors, Lean bounds the second derivative along a segment by $5(\delta_i^2 + \delta_{i+1}^2)$, giving the remainder in section 7 without individual Hessian bounds. The sixth-radius bound is also proved directly: for $0 < a, b < 2$, $a^2 + b^2 - ab \le \max(a^2, b^2) < 4$, so any separated pair has contact angle greater than $\pi/3$. Six such centers would require more than a full turn. This retains the feasible boundary case at radius 2 and avoids assuming the six-disk optimum. The exact wall radius is proved to lie in the refined certificate interval $(35213569647/10^{10}, 35213569648/10^{10})$ by rational square comparisons.

**Trust boundary.** The final theorem depends on finite replays using `native_decide`. Running `#print axioms` on the closed-form theorem reports the usual logical axioms and a family of generated native-evaluation axioms inherited from the Stage 0, Stage 3, Stage 4, and coverage replays. There are no `sorry` placeholders in this theorem chain. These native axioms mean the finite Boolean decisions are accepted using Lean's compiled evaluator; the trusted base therefore includes

the Lean compiler and code marked `@[implemented_by]`. This is faster than kernel reduction, but it is not a kernel-only proof. The Lean reference describes this tradeoff in its decision-procedure section.

The target and default umbrella module build with:

```
lake build CirclePacking.FifteenOptimality CirclePacking
```

To inspect the theorem's dependencies, place the following in an audit file and run `lake env lean Audit.lean` from the Lean directory:

```
import CirclePacking.FifteenOptimality
#print axioms CirclePacking.fifteen_unit_packing_radius_iff_ge_closed_form
```

## References


[1] R. L. Graham, "Sets of points with given minimum separation (Solution to Problem E1921)," *American Mathematical Monthly* 75 (1968), 192–193.

[2] U. Pirl, "Der Mindestabstand von $n$ in der Einheitskreisscheibe gelegenen Punkten," *Mathematische Nachrichten* 40 (1969), 111–124.

[3] R. L. Graham, B. D. Lubachevsky, K. J. Nurmela, and P. R. J. Ostergård, "Dense packings of congruent circles in a circle," *Discrete Mathematics* 181 (1998), 139–154. https://doi.org/10.1016/S0012-365X(97)00050-2.

[4] D. B. Ekanayake and D. J. LaFountain, "Tight partitions for packing circles in a circle," *Italian Journal of Pure and Applied Mathematics* 51 (2024), 115–136. https://ijpam.uniud.it/online_issue/202451/10%20Ekanayake-LaFountain.pdf.

[5] E. Specht, "The best known packings of equal circles in a circle," Packomania, https://www.packomania.com/cci/cci.html, accessed October 9, 2026.